Index Score

A tech-weighted composite regime indicator for TradingView (Pine Script v5)

Document purpose: explain what the Index Score Pine Script does, how it is calculated, what inputs it uses, how to interpret the output, and practical guidelines for when to run it.
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This indicator combines four market components into a single score: (1) a technology leadership signal, (2) a breadth/participation proxy, (3) the S&P; 500 trend itself, and (4) a volatility (risk) component. Each component is normalized (z-score), weighted, combined, then optionally scaled into an easy-to-read range such as 0-100.

Quick start	Add the indicator to a SPY chart (or your S&P; proxy). Use the 1D timeframe for regime, and optionally a lower timeframe (for example 1H or 4H) for tactical context. Treat the score as a filter, not a standalone entry signal.

1. Overview

Index Score is a single-number market regime indicator designed to reduce the risk of misreading the market during periods where technology-heavy mega caps dominate index movement. It intentionally gives technology leadership a configurable weight while still incorporating a participation proxy (breadth), the index trend, and a volatility/risk component.

The output is a continuous score. Higher values imply a more bullish, risk-on environment; lower values imply a more bearish, risk-off environment. Because the components are computed from external symbols (SPY, RSP, VIX, and selected tech symbols), the score is primarily a macro filter, not a stock selection tool.

2. What the script measures

The script is built from four component signals, each converted into a standardized z-score (so they can be combined):

Tech leadership (techZ): strength of technology relative to its own trend, using one of: a MegaCap 7 basket, XLK, QQQ, or a custom symbol.

Breadth/participation proxy (breadthZ): momentum of the ratio RSP/SPY (equal-weight vs cap-weight). Rising ratio implies broad participation.

Index trend (marketZ): momentum of SPY itself, to avoid overreacting to internals when price trend is clearly strong or weak.

Volatility / risk (volZ): inverse momentum of VIX. Falling VIX (risk-on) increases the score; rising VIX decreases it.

3. Core calculation (high level)

Each component is computed as a momentum series, then standardized via z-score, then combined with user weights:

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compositeZ = (wTech*techZ + wBreadth*breadthZ + wMarket*marketZ + wVol*volZ) /
(wTech+wBreadth+wMarket+wVol)
```

Finally, the composite z-score can be displayed as raw z, or scaled to -100 to +100, or 0 to 100. Scaling uses a tanh transform to keep the output bounded and readable.

4. Inputs and what they do

4.1 Data inputs

Input	Default	Purpose / effect
Component timeframe	blank (chart)	If set, all external symbols are sampled on this timeframe via request.security(). This lets you run the indicator on any chart while keeping a fixed regime timeframe.
S&P; 500 proxy	AMEX:SPY	Primary market series used for marketZ and in the breadth ratio.
Equal-weight proxy	AMEX:RSP	Used to build RSP/SPY for breadthZ.
Volatility proxy	CBOE:VIX	Used for volZ (inverted).

4.2 Tech component inputs

Tech leadership can be derived from a single ETF, a single symbol, or a weighted basket. Choose via Tech mode.

Tech mode	Description	Notes
MegaCap 7	Weighted basket of 7 symbols	Use inputs sym1..sym7 and weights w1..w7. Optionally auto-normalize weights.
XLK	Technology sector ETF	Simplest tech proxy with minimal external symbol requests.
QQQ	Nasdaq 100 ETF	Broader growth/tech proxy.
Custom	Single custom symbol	Use symTechCustom (for example XLK, QQQ, MSFT, NVDA, etc.).

4.3 Calculation inputs

Input	Default	Purpose / effect
Trend EMA length (trendLen)	50	Defines the baseline trend used by the 'EMA Distance' momentum mode.
Z-score length (zLen)	200	Lookback for mean and standard deviation used in z-score normalization. Larger values = slower, more stable normalization.
Smoothing EMA length (smoothLen)	5	Smooths each momentum series before z-scoring; higher values reduce noise but add lag.
Momentum source (momentumMode)	EMA Distance	EMA Distance uses (price/EMA(trendLen)-1). Log ROC uses log(price)-log(price[rocLen]).
Log ROC length (rocLen)	20	Used only if momentumMode = Log ROC; larger values measure longer-term rate of change.

4.4 Composite weights

Weights determine how much each component influences the final score. They are automatically normalized by the sum of weights.

Weight	Default	Effect
wTech	0.55	Higher increases sensitivity to tech leadership and concentration regimes.
wBreadth	0.25	Higher increases sensitivity to broad participation (equal-weight strength).
wMarket	0.10	Higher makes the score track SPY's own trend more closely.
wVol	0.10	Higher increases sensitivity to volatility regime shifts (risk-on/off).

5. Output, scaling, and visualization

The indicator plots a single series called Index Score. The same underlying composite is displayed in three alternative formats:

Raw (z): compositeZ (unbounded, typically around -3 to +3).

-100 to +100: 100 * tanh(compositeZ / scaleFac).

0-100: 50 * (tanh(compositeZ / scaleFac) + 1).

The tanh transform compresses extremes, which makes the oscillator more stable and readable. Use scaleFac to control sensitivity: smaller values make the output swing faster; larger values make it calmer.

5.1 Plot style and colors

You can render the score as a line, histogram, columns, area, step, line break, circles, or cross. The plotted series uses a gradient color: bearish (low), neutral (mid), bullish (high).

Display toggles include guide levels, component plots (z only), and a table that prints current values for each component and the composite.

6. Alerts

The script provides two alert conditions for regime shifts. Thresholds depend on the selected scale:

Scale	Bullish cross threshold	Bearish cross threshold
0-100	60	40
-100 to +100	20	-20
Raw (z)	0.5	-0.5

Practical use: set alerts on SPY (or your index chart) to notify you when the environment flips from risk-off to risk-on (and vice versa).

7. How and when to use Index Score

Index Score is designed as a regime filter. It is most useful when paired with your existing setup/entry method (breakouts, pullbacks, mean reversion, options structures, etc.).

Suggested workflows:

Daily regime (1D): use to set risk posture (aggressive vs selective vs defensive).
Tactical context (1H/4H): use to avoid fighting the tape intraday or on short swings.
Weekly regime (1W): use to prevent overreacting to daily noise.

Because the script is based on index-level symbols, running it as a screener across a watchlist generally provides little value: it will return roughly the same score regardless of the chart symbol. Use it on an index chart, and use a separate screener for stock selection.

8. What to expect from the output

When tech leadership is strong, volatility is benign, and participation is improving, the score should rise and stay elevated. When volatility rises and participation weakens (especially alongside a weakening SPY trend), the score should fall.

Common patterns:

High techZ, low breadthZ: concentration regime (leaders carry the index). Expect choppy/non-uniform performance across the average stock.
Rising breadthZ: broadening rally; more groups participate; trend-following tends to work better across more tickers.
Falling volZ (i.e., rising VIX): risk tightening; more whipsaws; consider smaller size or tighter criteria.

If your chart is on a very low timeframe and you keep the component timeframe blank, the score can become noisy. To keep it stable, set the Component timeframe to 60, 240, or 1D.

9. Limitations and notes

1) It is not a guarantee of future returns. It is a summary of recent conditions.
2) It relies on symbol data availability (SPY, RSP, VIX, and chosen tech symbols). If any symbol is unavailable on your plan or exchange feed, that component may degrade or return NA/0.
3) Z-score normalization depends on the lookback zLen; changing zLen changes what qualifies as 'extreme'.
4) The tech basket is user-editable. If market leadership changes over time, update the tickers/weights.

Appendix A: Key variables (names in the script)

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spy, rsp, vix (source series); marketZ, breadthZ, volZ, techZ (component z-scores);
compositeZ (weighted z-score); score (displayed output).
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